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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Course Outcome	4

Case Study Topic: Flash-Sale Order Dispatch System – Fast "Rank 5,00,000 Orders by Price, Priority & PIN Code" Query using Heap Sort, Counting Sort, and Radix Sort

Word Done Summary:

In this case study, three advanced sorting algorithms — Heap Sort, Counting Sort, and Radix Sort — were implemented to efficiently rank orders in a flash-sale dispatch system. The system handles approximately 5,00,000 orders, where each order carries a price, a priority level (1–10), and a 6-digit delivery PIN code. A normal comparison-based sort can rank any of these keys in O(n log n) time, but it does not exploit the bounded range of the priority and PIN-code fields. To improve query performance, Heap Sort was used to rank orders by price (an unbounded numeric key), Counting Sort was used to rank orders by priority level (a small bounded range), and Radix Sort was used to rank orders by PIN code (a fixed-width multi-digit key), each chosen to match the structure of its key.

Implementation Details:

1. Max-heap construction over order prices for Heap Sort.
2. Heapify and repeated root-extraction to produce the price-sorted order.
3. Frequency-array and prefix-sum construction for Counting Sort on priority levels.
4. Stable placement of orders into the output array during Counting Sort.
5. Digit-wise (LSD) passes over PIN codes for Radix Sort, using Counting Sort as a stable subroutine per digit.
6. Insertion of sample order records (price, priority, PIN code) into all three structures.
7. Sorted-output verification for each algorithm against expected order.
8. Complexity and stability analysis for each algorithm.

Result:

All order prices were sorted using Heap Sort, priority levels were sorted using Counting Sort, and PIN codes were sorted using Radix Sort. Stability was preserved in Counting Sort and Radix Sort, and all three sorted outputs were verified to be correct.

Output:

```
Heap Sort (Price):
499 899 1499 2499 3499 4999 7999 12499

Counting Sort (Priority 1-10):
1 1 2 2 3 5 7 9 10

Radix Sort (PIN Code):
100023 100089 100456 245112 500001 682001

Highest-priority order to dispatch first: Priority 1
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GitHub Link:

https://github.com/vigneshkumar-stack/DSA-Project

Student Signature -------------------	**Faculty Signature** -------------------